

PathoFMPred complete three-representation output

COAD TCGA-AA-A01F | TITAN, Giga-SSL and Prov-GigaPath | 2026-09-20

UNVALIDATED INTERNAL TCGA RESEARCH OUTPUT. BINARY REFERENCE RANK IS NOT PROBABILITY. DO NOT USE FOR DIAGNOSIS, TREATMENT SELECTION OR CLINICAL RISK ESTIMATION.

Output inventory

Representation	Continuous outputs	Binary outputs	Slides pooled	Output rule
TITAN	15	8	1	Q2 >= 0.20; AUROC >= 0.60
Giga-SSL	12	7	1	Q2 >= 0.20; AUROC >= 0.60
Prov-GigaPath	14	7	1	Q2 >= 0.20; AUROC >= 0.60

Case context, not used as model input

The sigmoid resection contained an ulcerated grade-2 adenocarcinoma that had crossed the muscular wall into pericolic fat (pT3). Both resection ends were free of carcinoma; lymphatic invasion was recorded, and 2 of 30 regional nodes contained metastasis (pN1).

Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3).

How to read the report

- PathoFMPred lists only representation-specific fitted objects that crossed the recorded internal performance threshold and passed the package's default sample-size rule.
- Continuous predictions remain in the source endpoint units. The internal TCGA rank provides display context and is not a clinical reference interval.
- Binary calls use an uncalibrated fitted LDA rule. The original LDA score is model-specific; neither the score nor its TCGA rank is a probability.
- The example patients lack reference labels for the displayed outcomes in the processed analysis table, so this report demonstrates software output rather than correctness.

Feature group	Reference label source
Mutations	TCGA MC3 protein-altering calls within the Bailey tissue-specific driver catalogue
Genome doubling and aneuploidy	Taylor et al. arm-level and genome-doubling variables
MSI	cBioPortal TCGA PanCancer MANTIS and MSIsensor variables
Oncogenic pathways	Sanchez-Vega pathway alteration matrix
Immune and genomic context	Published Thorsson PanImmune computational phenotypes

TITAN: complete predictable output

This page reports all 15 continuous and 8 binary outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score	8.211214	source endpoint units	25.4	0.302	0.575	408	Retained under code grouping
MSI	MANTIS score	0.448561	source endpoint units	61.5	0.284	0.4	391	Code-group sensitive: below original documented screening threshold
MSI	MSIsensor score	7.925266	source endpoint units	72.4	0.222	0.253	411	Code-group sensitive: below original documented screening threshold
Rna Pathway Activity	HALLMARK_ALLOGRAFT_REJECTION	-0.024224	source endpoint units	46.8	0.233	0.474	424	Code-group sensitive: below original documented screening threshold
Rna Pathway Activity	HALLMARK_IL6_JAK_STAT3_SIGNALING	0.074934	source endpoint units	63.8	0.206	0.47	424	Code-group sensitive: below original documented screening threshold
Rna Pathway Activity	HALLMARK_INTERFERON_ALPHA_RESPONSE	0.086532	source endpoint units	65.4	0.215	0.475	424	Code-group sensitive: below original documented screening threshold
Rna Pathway Activity	HALLMARK_INTERFERON_GAMMA_RESPONSE	0.068383	source endpoint units	63.1	0.251	0.507	424	Code-group sensitive: below original documented screening threshold
Rna Pathway Activity	HALLMARK_MYOGENESIS	-0.328593	source endpoint units	5.8	0.22	0.441	424	Code-group sensitive: below original documented screening threshold
Immune/context	Aneuploidy Score	6.658038	source endpoint units	24.1	0.217	0.518	408	Retained under code grouping
Immune/context	Lymphocyte Infiltration Signature Score	-0.40869	source endpoint units	25.1	0.297	0.532	417	Retained under code grouping
Immune/context	Macrophage Regulation	-0.51273	source endpoint units	35.1	0.226	0.471	417	Code-group sensitive: near chance or worse
Immune/context	Nonsilent Mutation Rate	1.814873	log1p-transformed source endpoint units	72.3	0.275	0.345	345	Code-group sensitive: near chance or worse
Immune/context	SNV Neoantigens	4.704855	log1p-transformed source endpoint units	77	0.241	0.363	350	Code-group sensitive: below original documented screening threshold
Immune/context	Silent Mutation Rate	1.096294	log1p-transformed source endpoint units	70.9	0.259	0.368	345	Code-group sensitive: near chance or worse
Immune/context	TIL Regional Fraction	9.860742	source endpoint units	56.4	0.407	0.69	414	Retained under code grouping

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC	PR-AUC	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Absent	-2.153	32.2	0.735	0.644	408 (165/243)	Retained under code grouping
Mutation	APC	Wild type	-2.982	17.7	0.851	0.822	382 (196/186)	Code-group sensitive: near chance or worse
Mutation	KRAS	Wild type	-5.333	12.2	0.759	0.559	382 (126/256)	Retained under code grouping
Mutation	TP53	Wild type	-3.171	17.4	0.852	0.789	382 (161/221)	Retained under code grouping
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-3.146	60.7	0.824	0.626	391 (74/317)	Retained under code grouping
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-2.097	82.2	0.902	0.714	374 (57/317)	Retained under code grouping
Pathway	MYC	Not altered	-1.27	61.5	0.647	0.375	323 (79/244)	Code-group sensitive: near chance or worse
Pathway	TP53	Altered	0.836	44.7	0.734	0.815	323 (208/115)	Retained under code grouping

AUROC and PR-AUC are internal repeated out-of-fold performance estimates. The class call shown here comes from the fitted model object and is not a calibrated probability. Grouped sensitivity uses TCGA tissue-source-site code, a barcode-derived cohort variable rather than an institution, scanner or laboratory identifier.

Giga-SSL: complete predictable output

This page reports all 12 continuous and 7 binary outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score	13.885722	source endpoint units	65.2	0.26	0.514	408	Grouped sensitivity not evaluated
Aneuploidy	Deleted arm count	6.163585	source endpoint units	46.8	0.209	0.528	408	Grouped sensitivity not evaluated
MSI	MANTIS score	0.306861	source endpoint units	17.3	0.266	0.44	391	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_ALLOGRAFT_REJECTION	-0.097504	source endpoint units	40.1	0.312	0.581	424	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_IL6_JAK_STAT3_SIGNALING	0.05774	source endpoint units	60.9	0.245	0.515	424	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_INTERFERON_GAMMA_RESPONSE	-0.136539	source endpoint units	37.7	0.277	0.543	424	Grouped sensitivity not evaluated
Immune/context	Aneuploidy Score	12.052227	source endpoint units	68.9	0.238	0.503	408	Grouped sensitivity not evaluated
Immune/context	Leukocyte Fraction	0.181064	source endpoint units	52.8	0.288	0.557	419	Grouped sensitivity not evaluated
Immune/context	Lymphocyte Infiltration Signature Score	-0.230227	source endpoint units	39.5	0.273	0.516	417	Grouped sensitivity not evaluated
Immune/context	Macrophage Regulation	-0.588178	source endpoint units	32.3	0.284	0.548	417	Grouped sensitivity not evaluated
Immune/context	SNV Neoantigens	4.261001	log1p-transformed source endpoint units	49.9	0.226	0.361	350	Grouped sensitivity not evaluated
Immune/context	TIL Regional Fraction	8.853776	source endpoint units	52.2	0.377	0.628	414	Grouped sensitivity not evaluated

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC	PR-AUC	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Present	0.029	64.7	0.763	0.676	408 (165/243)	Grouped sensitivity not evaluated
Mutation	APC	Wild type	-3.817	17.7	0.795	0.73	382 (196/186)	Grouped sensitivity not evaluated
Mutation	KRAS	Mutation	-2.143	24.7	0.693	0.457	382 (126/256)	Grouped sensitivity not evaluated
Mutation	TP53	Wild type	-3.697	14	0.786	0.673	382 (161/221)	Grouped sensitivity not evaluated
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-3.919	65.3	0.777	0.533	391 (74/317)	Grouped sensitivity not evaluated
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-5.615	62.3	0.851	0.584	374 (57/317)	Grouped sensitivity not evaluated
Pathway	TP53	Not altered	-1.099	15	0.749	0.842	323 (208/115)	Grouped sensitivity not evaluated

AUROC and PR-AUC are internal repeated out-of-fold performance estimates. The class call shown here comes from the fitted model object and is not a calibrated probability. Grouped sensitivity uses TCGA tissue-source-site code, a barcode-derived cohort variable rather than an institution, scanner or laboratory identifier.

Prov-GigaPath: complete predictable output

This page reports all 14 continuous and 7 binary outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score	13.905074	source endpoint units	63.8	0.294	0.561	407	Grouped sensitivity not evaluated
Aneuploidy	Deleted arm count	6.412989	source endpoint units	50.5	0.205	0.539	407	Grouped sensitivity not evaluated
MSI	MANTIS score	0.365703	source endpoint units	39.6	0.3	0.458	390	Grouped sensitivity not evaluated
MSI	MSIsensor score	2.298627	source endpoint units	39.6	0.254	0.327	410	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_ALLOGRAFT_REJECTION	-0.695598	source endpoint units	2	0.318	0.581	423	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_IL6_JAK_STAT3_SIGNALING	-0.282125	source endpoint units	17.1	0.244	0.533	423	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_INTERFERON_ALPHA_RESPONSE	-0.338695	source endpoint units	16.9	0.231	0.486	423	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_INTERFERON_GAMMA_RESPONSE	-0.432654	source endpoint units	9.3	0.284	0.566	423	Grouped sensitivity not evaluated
Immune/context	Aneuploidy Score	11.692715	source endpoint units	66	0.296	0.562	407	Grouped sensitivity not evaluated
Immune/context	Leukocyte Fraction	0.151881	source endpoint units	32.2	0.274	0.538	418	Grouped sensitivity not evaluated
Immune/context	Lymphocyte Infiltration Signature Score	-1.876459	source endpoint units	0.1	0.311	0.565	416	Grouped sensitivity not evaluated
Immune/context	Macrophage Regulation	-1.56487	source endpoint units	0.4	0.291	0.553	416	Grouped sensitivity not evaluated
Immune/context	SNV Neoantigens	4.248818	log1p-transformed source endpoint units	49.3	0.271	0.356	350	Grouped sensitivity not evaluated
Immune/context	TIL Regional Fraction	2.938463	source endpoint units	19.5	0.507	0.689	413	Grouped sensitivity not evaluated

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC	PR-AUC	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Absent	-0.515	52.2	0.765	0.701	407 (165/242)	Grouped sensitivity not evaluated
Mutation	APC	Wild type	-9.439	0.1	0.842	0.796	382 (196/186)	Grouped sensitivity not evaluated
Mutation	KRAS	Wild type	-3.535	20	0.74	0.526	382 (126/256)	Grouped sensitivity not evaluated
Mutation	TP53	Wild type	-4.388	9.8	0.787	0.695	382 (161/221)	Grouped sensitivity not evaluated
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-3.124	61.7	0.822	0.618	390 (73/317)	Grouped sensitivity not evaluated
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-6.191	51.3	0.857	0.62	373 (56/317)	Grouped sensitivity not evaluated
Pathway	TP53	Altered	2.33	66.4	0.724	0.809	323 (208/115)	Grouped sensitivity not evaluated

AUROC and PR-AUC are internal repeated out-of-fold performance estimates. The class call shown here comes from the fitted model object and is not a calibrated probability. Grouped sensitivity uses TCGA tissue-source-site code, a barcode-derived cohort variable rather than an institution, scanner or laboratory identifier.